

Assignment module 6: Network Security, Maintenance, and Troubleshooting
Procedures

Section 1: Multiple Choice

1. What is the primary purpose of a firewall in a network security infrastructure?

- a) Encrypting network traffic
- b) Filtering and controlling network traffic
- c) Assigning IP addresses to devices
- d) Authenticating users for network access

ANS - B

2. What type of attack involves flooding a network with excessive traffic to disrupt normal operation?

- a) Denial of Service (DoS)
- b) Phishing
- c) Spoofing
- d) Man-in-the-Middle (MitM)

ANS - A

3. Which encryption protocol is commonly used to secure wireless network communications?

- a) WEP (Wired Equivalent Privacy)
- b) WPA (Wi-Fi Protected Access)
- c) SSL/TLS (Secure Sockets Layer/Transport Layer Security)
- d) AES (Advanced Encryption Standard)

ANS - B

4. What is the purpose of a VPN (Virtual Private Network) in a network security context?

ANS – The primary function of vpn is to hide the original ip of device and create a virtual ip and show it to insure private and secure communication in public.

True or False: Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.

ANS - TRUE

True or False: A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.

ANS - TRUE

True or False: Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

ANS - TRUE

8. Describe the steps involved in conducting a network vulnerability Assignment.

ANS –

- 1. Define The Objective**
- 2. Watch For the Network type like its topology and structure.**

3. **Scan For the Vulnerability.**
4. **Check all port are connected properly.**
5. **Check for nic card.**
6. **Try ping command and check for system is up or down .**
7. **If there is problem in your pc check nic card or install new.**

10. Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

ANS -

Improves Network Perfomance and Stability.

Enhances Security and Detects for the future possible attacks and protect from it.

Protects Data From loss by Tacking Back up from it.

1. **Regular software Updates.**
2. **Regularly Watching network Performance.**
3. **Taking Backups.**
4. **Checking hardware health.**
5. **Checking for the security of network.**
6. **Regularly testing network connectivity.**

Thos all are the key points .

1. Which of the following best describes the purpose of a VPN (Virtual Private Network)?

- a) Encrypting network traffic to prevent eavesdropping
- b) Connecting multiple LANs (Local Area Networks) over a wide area network (WAN)
- c) Authenticating users and controlling access to network resources
- d) Reducing latency and improving network performance

ANS - A